

Ahmed Essam Abd Elgawad

Senior Data Scientist | AI & Machine Learning Engineer

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Work Experience

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Senior Data Scientist

June 2025 – Present

Developed advanced NLU systems for intent classification and entity recognition using Azure CLU and custom LargeRoBERTa models, achieving superior accuracy in domain-specific language understanding. Designed and deployed enterprise-grade conversational AI pipelines leveraging Azure Cognitive Services for real-time production applications.

BG GROUP

Senior Artificial Intelligence and Machine learning engineer

Aug 2024 – May 2025

I developed predictive models for cancer risk analysis, patient clustering, and treatment recommendation. I also created HR- focused tools, including a predictive analytics system for employee retention and an NLP chatbot for HR automation.

Additionally, I implemented computer vision for cancer diagnostics and led the development of a secure and liveness-enabled ID verification system for attendance tracking.

Micro1 AI

Data Scientist

Mar 2024 – Aug 2024

I worked as a Data Scientist to design and develop prompts across various software engineering domains to train advanced LLMs like GPT-3. I created real-world coding challenges to improve the models' understanding of programming tasks and provided high-quality, structured training data. This role strengthened my expertise in prompt engineering, enabling me to guide LLMs toward precise functionalities and solutions.

Hassan Allam Holding

Data Scientist

Jan 2021 – Dec 2022

I led a time series forecasting project to predict monthly material needs, using ARIMA and LSTM models for accuracy. This project optimized resource allocation and achieved significant cost savings by reducing excess inventory. I collaborated with cross-functional teams to turn data insights into strategies that ensured seamless project continuity and effective material management.

Education

Master of Engineering in Artificial Intelligence and Data Science

Faculty of Engineering, University of Ottawa | Canada

Jan 2023 – Jan 2024

- **Courses:** Natural Language Processing, Computer Vision, Machine Learning, Data Science, AI for CS, Cloud Computing

Bachelor of Engineering in Electronic and Communication Engineering

Alexandria University | faculty of engineering

Jan 2014 – Oct 2019

- **Grade:** Very Good | **GPA:** 3.32 / 4.00
- **Graduation Project:** Design and implementation of an automated detection system

Technical Skills

- **Programming language:** Python, C++, Swift, SQL, Git, Unit Testing, OOP, Design Patterns, familiar with (Java, R)
- **Machine Learning & AI:** Supervised and Unsupervised Learning (XGBoost, Gradient boosting, Random Forest, SVM, k-Means, PCA), Deep Learning (PyTorch, TensorFlow, RNN, LSTM), Transformers (BERT, RoBERTa, AraBERT, mBERT, MT5, ALLaM), Large Language Models (LLMs), Generative AI (Hugging Face, IBM Watsonx, OpenAI), Retrieval-Augmented Generation (RAG), Ensemble Learning, Agentic AI.
- **Natural Language Processing & Data Processing:** Text Classification, Sentiment Analysis, Multilingual and Arabic NLP, Vector Databases (Weaviate, ChromaDB), Word embedding (GloVe, Word2Vec, Tf-Idf), Data Cleaning, Feature Engineering, Pandas, NumPy, NLTK, SpaCy, Regular Expressions (Regex).
- **Cloud & Data Engineering:** Azure (AI Fundamentals, Data Factory), SQL-based solutions, ETL, NoSQL (MongoDB), familiar with Big Data ecosystems (Hadoop, Spark).
- **Business Intelligence & Visualization:** Power BI, Tableau, Matplotlib, Seaborn, Teradata, SAP.
- **Statistical Analysis & Optimization:** Hypothesis Testing, Regression Analysis, Hyperparameter Tuning (Grid Search).

Projects

Audio Deepfake Detection

[GitHub](#)

The project aimed to develop an efficient and accurate audio deepfake detection system to identify manipulated audio recordings, mitigating the spread of misinformation and preserving the integrity of audio content.

Designing a Combination Scheme of Learning Methods for Botnet Detection

[GitHub](#)

Developed a machine learning-based botnet detection system to enhance cybersecurity for IoT networks, focusing on anomaly detection and feature selection techniques to differentiate between benign and malicious traffic data.

Stock Market Prediction using Sentiment Analysis

[GitHub](#)

Enhancing stock market prediction by integrating Twitter sentiment analysis due to the unpredictable nature of stock markets, affected by various factors like social media trends and political variables.

Rankyx AI Procurement System

[GitHub](#)

A multi-agent AI system powered by CrewAI that automates product research and procurement analysis. The system uses intelligent agents to search, scrape, and analyze product data from multiple e-commerce websites, generating comprehensive procurement reports.

Face Recognition and Anti-Spoofing System

[GitHub](#)

This project is a face recognition and anti-spoofing system built with Python, and OpenCV. It allows for secure and authenticated access via real-time face recognition, with liveness detection to prevent spoofing attempts. Users can log in, log out, and register new accounts.

Certifications

- **Cloud & AI Fundamentals**
 - [Microsoft Azure AI Fundamentals](#) – Microsoft, October 2023
 - [Microsoft Azure Fundamentals](#) – Microsoft, December 2023
 - [AWS Introduction to Generative AI](#) – Amazon Web Services, January 2024
- **Data Science & Machine Learning**
 - [Predictive Analytics Modeler](#) – IBM, February 2023
 - [Big Data Engineer](#) – IBM, March 2023
 - [Artificial Intelligence Analyst](#) – IBM, April 2023
- **Professional & Language Skills**
 - [Soft Skills Program](#) – Dale Carnegie, February - May 2023
 - [English Training](#) – Berlitz, February - May 2023